

## Solution Requirements

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Date</b>          | 06 July 2026                                      |
| <b>Team ID</b>       |                                                   |
| <b>Project Name</b>  | Smart Lender - Loan Eligibility Prediction System |
| <b>Maximum Marks</b> | 4 Marks                                           |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                                                    | Priority (High/Medium/Low) |
|------|----------------------|------------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | <i>Users securely register and log in using email/mobile and password.</i>                                 | High                       |
| 2    | Authorization levels | <i>Admin manages users and applications; applicants can submit and track loan applications.</i>            | High                       |
| 3    | External interfaces  | <i>Connects to Firebase/MySQL database, Email/SMS service, and optional Aadhaar/PAN verification APIs.</i> | Medium                     |

|   |                                   |                                                                                                                                    |        |
|---|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|
|   |                                   |                                                                                                                                    |        |
| 4 | Transactions processing           | <i>Accepts loan applications, validates data, predicts eligibility using the ML model, and stores results.</i>                     | High   |
| 5 | Reporting                         | <i>Generates dashboards, loan eligibility reports, application history, and PDF/Excel exports.</i>                                 | Medium |
| 6 | Business rules, etc.              | <i>Applicant must be 18+ years, all mandatory fields completed, and eligibility calculated by the ML model.</i><br><br><i>High</i> | High   |
| 7 | Compliance to laws or regulations | <i>Protects user data, uses secure authentication, and follows Indian data privacy guidelines.</i>                                 | High   |
| 8 | <i>Other</i>                      | <i>Password reset, notifications, search/filter options, and audit logs.</i>                                                       | Medium |

**Step 2: Non-Functional Requirements (NFR)**

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|      |              |                         |                                     |

|          |                            |                                                                                                                      |                                                |
|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>1</b> | Performance & Speed        | <i>Fast response for login and loan prediction.</i>                                                                  | <i>Page loads in &lt; 2 seconds</i>            |
| <b>2</b> | Scalability                | <i>Supports increasing users and loan records.</i>                                                                   | <i>5,000+ users</i>                            |
| <b>3</b> | Security & Data Privacy    | <i>Encrypt user data and use secure authentication.</i>                                                              | <i>HTTPS + AES-256 encryption.</i>             |
| <b>4</b> | Reliability & Availability | <i>System should be available continuously.</i>                                                                      | <i>99.9% uptime</i>                            |
| <b>5</b> | Usability & Accessibility  | <i>User-friendly interface that works on desktop and mobile.</i><br><br><i>Easy navigation and responsive design</i> | <i>Easy navigation and responsive design.</i>  |
| <b>6</b> | <i>Other</i>               | <i>Easy maintenance with regular backups and recovery.</i>                                                           | <i>Daily backup and easy software updates.</i> |